

**SENIOR SOFTWARE &
DATA SCIENTIST |
SEMICONDUCTOR
PROCESS MODELING |
PH.D. PHYSICIST**

SKILLS

PROGRAMMING

Python / C++ / MATLAB

DATA SCIENCE

Machine Learning, Data
Analysis, Predictive Modeling,
Experimental Design

TOOLS

Pandas, NumPy, Keras,
Jupyter, Git

PROJECT

MANAGEMENT

Process/Product
Development, Cross-
Functional Collaboration

SEMICONDUCTOR

EXPERTISE

OPC models, RET,
Etch & Lithography
Process Modeling

RESEARCH

COMMUNICATIONS

Scientific Writing,
Conference Presentation

Min-Cheol Lee

Experienced engineer and data scientist with a strong background in semiconductor process modeling, photolithography, and data-driven yield optimization. Adept at building scalable machine learning models, automating workflows, and solving complex problems in advanced manufacturing environments. Ph.D. in Condensed Matter Physics with deep expertise in big data analysis and optical spectroscopy.

WORK EXPERIENCE

SOFTWARE RESEARCH ENGINEER / SCIENTIST

Intel Corporation, OR, USA

2022 – present

- Led OPC model development for 20+ layers on Intel 18A/14A nodes, collaborating with over 100 photolithography, etch, and device engineers.
- Integrated OPC models with industry leading photolithography techniques, leveraging both low-NA and high-NA EUV and DUV systems.
- Developed machine learning based OPC models trained on fab data, achieving a 20-50% reduction in yield variance across photomask products.
- Analyzed over 10,000 SEM images from Intel 18A and 14A nodes to identify and resolve yield and performance bottlenecks, pinpointing key photolithography failure mechanisms.
- Developed automated workflows for test photomask design and validation using Python, C++ and EDA software, reducing manual design error validation cycle from 1 week to 1-2 days, a 70 % cycle time improvement.

POSTDOCTORAL RESEARCHER

Los Alamos National Laboratory, NM, USA

2019 – 2022

- Led and participated ultrafast optical and X-ray spectroscopy experiments at SLAC/SACLA/PAL and other labs across 10+ projects, collaborating 50+ scientists.
- Analyzed terabyte-scale temporal/spectral datasets to reveal non-equilibrium dynamics in quantum materials from superconductors to topological systems, resulting in 10+ peer-reviewed publications in Nature Materials, Nature Communications, and Physical Review Letters.

EDUCATION

SEOUL NATIONAL UNIVERSITY

Doctor of Physics, Condensed Matter Physics

2013 – 2019

- Dissertation: Non-equilibrium Spectroscopic Studies on Coherent Phonon Oscillations

SEOUL NATIONAL UNIVERSITY

Bachelor's degree, Physics

2009 – 2013

PUBLICATIONS

- 26 publications | first author in 8 | cited over 600 times | [Google Scholar](#)
- Journals include Nature Materials, Physical Review Letters, Advanced Materials
- Selected publications:
 - [Phys. Rev. Lett. 128, 155301 \(2022\)](#) | [Communications Physics 5, 335 \(2022\)](#)
 - [Advanced Materials, 1704777 \(2018\)](#) | [Nature Materials 21, 62 \(2022\)](#)